SMU Lyle School of Engineering

Reinforcement Learning

Policy Gradient Methods

Sutton/Barto* Chapter 13

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With figures from Sutton/Barto*

*Sutton and Barto, Reinforcement Learning: An Introduction, 2nd edition, MIT Press, Cambridge, MA, 2018

Online Material

Topics of this Course

- Introduction to reinforcement learning
- Markov decision processes
- Part I: Tabular Methods
  - Dynamic programming
  - Monte Carlo methods
  - Temporal-difference learning
  - Multi-step bootstrapping
  - Planning and learning with tabular methods
- **Part II: Approximate Solution Methods**
  - Prediction and Control using Approximation
  - Eligibility Traces
  - **Policy Gradient Methods**
- Part III: Modern RL Methods
  - Deep Reinforcement Learning
  - Current Applications

Summary of Notation

General

X	capital letters: random variables
x, p	lower-case letters: realizations of random variables or scalar functions
$\boldsymbol{w}$	Bold lower-case letters: real-valued vectors (even if random variables)
$\boldsymbol{W}$	bold capitals: matrices
α	Greek letters: parameters (vectors if in bolt)
$\Pr\{X = x\}$	probability that a random variable X takes on the value x
$X \sim p$	random variable X selected from distribution $p(x) = \Pr\{X = x\}$
$\mathbb{E}[X]$	expectation of a random variable X, i.e., $\mathbb{E}[X] = \sum_x p(x)x$
$\text{argmax}_a f(a)$	a value of action a at which $f(a)$ takes its maximal value

Value Function

G_t	return (cumulative reward) following time t
$G_{t:h}$	return from t to h (discounted and corrected)
$v_\pi(s)$	value of state s under policy π (expected return)
$v_*(s)$	value of state s under the optimal policy
$q_\pi(s, a)$	value of taking action a in state s under policy π
$q_*(s, a)$	value of taking action a in state s under the optimal policy
V, V_t	array estimates of state-value function v_π or v_*
Q, Q_t	array estimates of action-value function q_π or v_*

Temporal Difference Learning

U_t	target for estimate at time t
δ_t	TD error

MDP

s, s'	states
a	an action
r	a reward
$\mathcal{S}$	set of all (nonterminal) states, $\mathcal{S}^+$ are all states
$\mathcal{A}(s)$	set of all actions available in state s
γ	discount-rate parameter
t	discrete time step
T	final time step of an episode (a.k.a. horizon)
A_t	random variable for the action at time t
S_t	random variable for the state at time t
R_t	random variable for the reward at time t
$p(s', r \mid s, a)$	probability of transition to state s' and receiving reward r, from state s taking action a.
$p(s' \mid s, a)$	probability of transition to state s' fom state s taking action a.
$r(s, a)$	expected immediate reward from state s after action a.
$r(s, a, s')$	expected immediate reward from state s to s' with action a.
$\pi(a \mid s)$	probability of taking action a in state s under stochastic policy π
$\pi(s)$	action taken in state s under deterministic policy π

Approximation

$\mu(s)$	on policy distribution (fraction of time spent in state)
α	step size (gradient descent learning rate)
$\hat{v}(s, \boldsymbol{w})$	approximate value function
$\hat{q}(s, a, \boldsymbol{w})$	approximate action-value function
$\pi(a \mid s, \boldsymbol{\theta})$	approximate policy
$\widehat{VE}$	mean squared value error
$\nabla f(x)$	gradient of function f evaluated at x

Parametrized Policies

Learn policies directly

Learn a Parameterized Policy

- **Up to now**: Learn parameters for an approximate action-value function $\hat{q}$ and then extract a greedy policy.

- **New idea**: Use a policy-based method that learn parameters for a policy.

  Parameterized stochastic policy:

$$\pi(a|s, \boldsymbol{\theta}) = \Pr\{A\_t = a \mid S\_t = s, \boldsymbol{\theta}\_t = \boldsymbol{\theta}\}, \qquad \boldsymbol{\theta} \in \mathbb{R}^d$$

- Learn $\boldsymbol{\theta}$ given a performance measure $J(\boldsymbol{\theta})$ using **approximate gradient ascent**.

  Update rule:

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \alpha \widehat{\nabla J(\boldsymbol{\theta}_t)}$$

  where $\widehat{\nabla J(\boldsymbol{\theta}_t)} \in \mathbb{R}^d$ is a stochastic estimate of the gradient of the performance measure with respect to $\boldsymbol{\theta}$. We use an estimate because getting the exact gradient is hard.

Policy Gradient Theorem

- We define the performance measure as the true value of following π_θ from the start state s_0

$$J(\boldsymbol{\theta}) \stackrel{\text{def}}{=} v_{\pi_\theta}(s_0) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{\infty} \gamma^t R_t \mid S_0 = s_0\right]$$

- The **policy gradient theorem** gives us an analytical expression for the unbiased gradient of the performance with respect to the policy parameters:

$$\nabla J(\boldsymbol{\theta}) = \mathbb{E}_\pi[q_\pi(s,a)\nabla \log \pi(a|s)] \propto \sum_s \mu(s) \sum_a q_\pi(s,a)\nabla\pi(a|s,\boldsymbol{\theta})$$

On-policy distribution under π

Depends on estimates for q

Gradient of the policy approximation function.

- **Convergence?**
  - Does **NOT** guarantee: convergence to (global) optimum, fast convergence or stability!

Parametrized Policy Example

Requirements

- $\pi(a|s,\boldsymbol{\theta})$ needs to be **differentiable** with respect to $\boldsymbol{\theta}$ so that $\nabla\pi(a|s,\boldsymbol{\theta})$ exists and is finite.
- π needs to stay soft to guarantee **exploration** and convergence.

Popular for a small, discrete action space: **Soft-max in action preferences**

1. Define a **numerical preference** function where you prefer state-action combinations with larger scores.

   $$h(s,a,\boldsymbol{\theta}) \in \mathbb{R}$$

   Some options:
   - Linear approximation with state (transformed) features $h(s,a,\boldsymbol{\theta}) = \boldsymbol{\theta}^{\top}\boldsymbol{x}(s,a)$
   - ANN with $\boldsymbol{\theta}$ as network weights.

2. Giving the action with the highest preference in each state the highest probability is done using **soft-max**:

   $$\pi(a|s,\boldsymbol{\theta}) = \frac{e^{h(s,a,\boldsymbol{\theta})}}{\sum_b e^{h(s,b,\boldsymbol{\theta})}}$$

   Note: Soft-max converts a set of scores into a probability distribution.

3. **Gradient**: $\nabla \ln \pi(s|a,\boldsymbol{\theta}) = \boldsymbol{x}(s,a) - \sum_b \pi(b|s,\boldsymbol{\theta})\boldsymbol{x}(s,b)$

Advantages Over ϵ-Greedy Action Selection

- **Learns stochastic policies**
  - The **stochastic policy will be driven to the optimal policy** which can also approach a deterministic policy.
  - Stochastic policies **perform exploration implicitly**. The amount of exploration is automatically reduced when it converges to the best action.

    Note: value-action methods cannot learn stochastic policies and are often stuck with learning an ϵ-greedy policy!

- **Smooth updates**
  - Parameterized policy changes action probability smoothly while ϵ-greedy policies jump when a different action becomes the maximum.

- **Incorporating prior knowledge is easier**
  - Prior knowledge about the form of the policy can be incorporated into the way the policy is parameterized (i.e., the approximation can be biased). This can improve learning significantly.

REINFORCE

Monte Carlo Policy Gradient Control

REINFORCE Update

Normal stochastic gradient-ascent algorithm update: $\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \alpha \nabla J(\boldsymbol{\theta}_t)$

Policy Gradient Theorem

$$\nabla J(\boldsymbol{\theta}) \propto \sum_s \mu(s) \sum_a q_\pi(s,a) \nabla \pi(a|s,\boldsymbol{\theta})$$

(replace s by the sampled $S_t \sim \mu$)

$$= \mathbb{E}_\pi \left[\sum_a \pi(A_t|S_t,\boldsymbol{\theta}) q_\pi(S_t,a) \frac{\nabla \pi(A_t|S_t,\boldsymbol{\theta})}{\pi(A_t|S_t,\boldsymbol{\theta})} \right]$$

(replace a by the sampled $A_t \sim \pi$)

$$= \mathbb{E}_\pi \left[G_t \frac{\nabla \pi(A_t|S_t,\boldsymbol{\theta})}{\pi(A_t|S_t,\boldsymbol{\theta})} \right]$$

(because $\mathbb{E}_\pi[G_t|S_t,A_t] = q_\pi(S_t,A_t)$)

$$= \mathbb{E}_\pi[G_t \ln \nabla \pi(A_t|S_t,\boldsymbol{\theta})]$$

(because $\nabla \ln x = \frac{\nabla x}{x}$)

A_t, S_t and an unbiased estimate of G_t under the current π can be obtained via MC.

Convergence to a local optimum:
- linear approximation
- step size reduction

REINFORCE Algorithm

REINFORCE: Monte-Carlo Policy-Gradient Control (episodic) for π_*

Input: a differentiable policy parameterization $\pi(a|s, \boldsymbol{\theta})$
Algorithm parameter: step size $\alpha > 0$
Initialize policy parameter $\boldsymbol{\theta} \in \mathbb{R}^{d'}$ (e.g., to $\mathbf{0}$)

Loop forever (for each episode):
  Generate an episode $S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T$, following $\pi(\cdot|\cdot, \boldsymbol{\theta})$
  Loop for each step of the episode $t = 0, 1, \ldots, T-1$:
    $G \leftarrow \sum_{k=t+1}^{T} \gamma^{k-t-1} R_k$ (G_t)
    $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha \gamma^t G \nabla \ln \pi(A_t|S_t, \boldsymbol{\theta})$

Notes:

- MC sampling leads to high variance in the gradient estimate (which is worse than variance in the return estimate).
- REINFORCE is often implemented with a baseline reward for each state to reduce variance: $R_t - b(S_t)$

Actor-Critic Methods

Combining an

- actor (a parameterized policy) with a
- critic (using a parameterized value functions)

Actor-Critic Methods

- **Idea**: estimate

  parameterized policy + parameterized value function

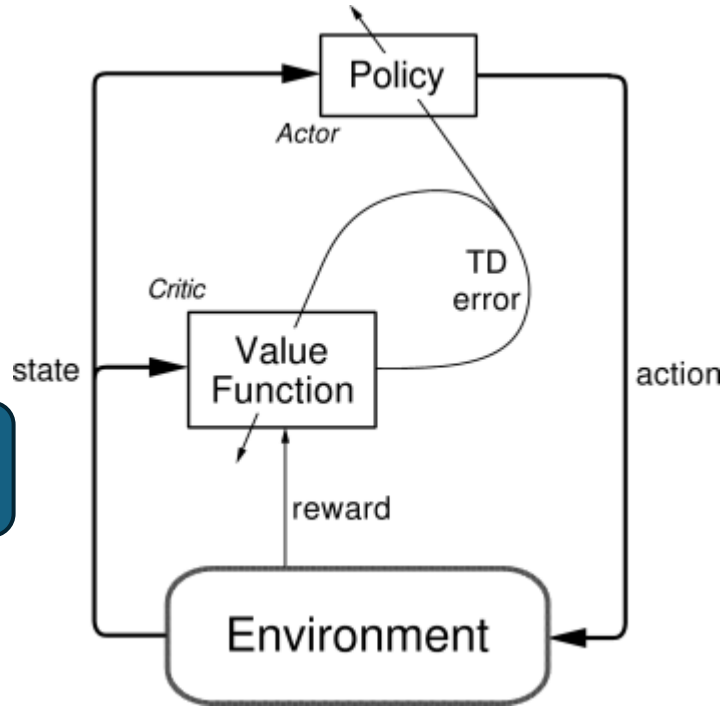
- **Actor**: estimates a parameterized policy π and decides on actions.
- **Critic**: estimates the TD error using its value function estimate $V = \hat{v}_\pi$

$$\delta_t = R_{t+1} + \gamma V(S_{t+1}) - V(S_t)$$

Can be seen as a learned baseline for the policy update!

The critic judges the actor's action by comparing the received reward to the expected value difference.

- Positive TD error: the last action performed better than expected and the tendency to pick it should be strengthened.
- Negative TD error: suggests the tendency should be weakened.



- The TD error is used to update both approximations! For the policy update this means:
  - The update does no longer use MC as in REINFORCE.
  - It uses a TD estimate of the return minus a value-function estimate as the baseline.
  - This reduces variability and can improve learning.

The Policy Gradient Update Signal

- True policy gradient

$$\nabla J(\boldsymbol{\theta}) = \mathbb{E}_\pi[q_\pi(s,a)\,\nabla \log \pi(a|s)]$$

- The update can use any signal X

$$\nabla J(\boldsymbol{\theta}) \propto \mathbb{E}_\pi[X\,\nabla \log \pi(a|s)]$$

  as long as $\mathbb{E}_\pi[X|s,a] \propto q_\pi(s,a)$, i.e., if it tells us how good an action is.

- Examples of valid signals:
  - MC (REINFORCE): $X = G_t$
  - Advantage function: $X = A(s,a) = q(s,a) - v(s)$
  - TD error: $X = \delta_t = R_{t+1} + \gamma v(S_{t+1}) - v(S_t)$

Advantage: Measures how much better a is compared with the current action in $\pi(s)$.

One-Step Actor-Critic Algorithm

One-step Actor–Critic (episodic), for estimating $\pi_{\boldsymbol{\theta}} \approx \pi_*$

Input: a differentiable policy parameterization $\pi(a|s,\boldsymbol{\theta})$

Input: a differentiable state-value function parameterization $\hat{v}(s,\mathbf{w})$

Parameters: step sizes $\alpha^{\boldsymbol{\theta}} > 0$, $\alpha^{\mathbf{w}} > 0$

Initialize policy parameter $\boldsymbol{\theta} \in \mathbb{R}^{d'}$ and state-value weights $\mathbf{w} \in \mathbb{R}^{d}$ (e.g., to $\mathbf{0}$)

Loop forever (for each episode):

  Initialize S (first state of episode)

  $I \leftarrow 1$

  Loop while S is not terminal (for each time step):

    $A \sim \pi(\cdot|S,\boldsymbol{\theta})$

    Take action A, observe S', R

    $\delta \leftarrow R + \gamma \hat{v}(S',\mathbf{w}) - \hat{v}(S,\mathbf{w})$ (if S' is terminal, then $\hat{v}(S',\mathbf{w}) \doteq 0$)

    $\mathbf{w} \leftarrow \mathbf{w} + \alpha^{\mathbf{w}} \delta \nabla \hat{v}(S,\mathbf{w})$

    $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha^{\boldsymbol{\theta}} I \delta \nabla \ln \pi(A|S,\boldsymbol{\theta})$

    $I \leftarrow \gamma I$

    $S \leftarrow S'$

actor

critic

Discounting multiplier: $\mathrm{I} = \gamma^t$

Policy approximation is soft, so it explores!

TD error: using $\hat{v}$ estimates.

Update the critic (value function) = $TD(0)$

Update the actor (policy) using the TD error signal.

Actor-Critic Methods with Eligibility Traces

One-step Actor–Critic (episodi

Input: a differentiable policy param
Input: a differentiable state-value fu
Parameters: step sizes $\alpha^{\boldsymbol{\theta}} > 0$, $\alpha^{\mathbf{w}} >$
Initialize policy parameter $\boldsymbol{\theta} \in \mathbb{R}^{d'}$
Loop forever (for each episode):
  Initialize S (first state of epis
  $I \leftarrow 1$
  Loop while S is not terminal (for
    $A \sim \pi(\cdot|S,\boldsymbol{\theta})$
    Take action A, observe S', R
    $\delta \leftarrow R + \gamma\hat{v}(S',\mathbf{w}) - \hat{v}(S,\mathbf{w})$
    $\mathbf{w} \leftarrow \mathbf{w} + \alpha^{\mathbf{w}}\delta\nabla\hat{v}(S,\mathbf{w})$
    $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha^{\boldsymbol{\theta}} I\delta\nabla\ln\pi(A|S,\boldsymbol{\theta})$
    $I \leftarrow \gamma I$
    $S \leftarrow S'$

Actor–Critic with Eligibility Traces (episodic), for estimating $\pi_{\boldsymbol{\theta}} \approx \pi_*$

Input: a differentiable policy parameterization $\pi(a|s,\boldsymbol{\theta})$
Input: a differentiable state-value function parameterization $\hat{v}(s,\mathbf{w})$
Parameters: trace-decay rates $\lambda^{\boldsymbol{\theta}} \in [0,1]$, $\lambda^{\mathbf{w}} \in [0,1]$; step sizes $\alpha^{\boldsymbol{\theta}} > 0$, $\alpha^{\mathbf{w}} > 0$
Initialize policy parameter $\boldsymbol{\theta} \in \mathbb{R}^{d'}$ and state-value weights $\mathbf{w} \in \mathbb{R}^{d}$ (e.g., to $\mathbf{0}$)
Loop forever (for each episode):
  Initialize S (first state of episode)
  $\mathbf{z}^{\boldsymbol{\theta}} \leftarrow \mathbf{0}$ (d'-component eligibility trace vector)
  $\mathbf{z}^{\mathbf{w}} \leftarrow \mathbf{0}$ (d-component eligibility trace vector)
  $I \leftarrow 1$
  Loop while S is not terminal (for each time step):
    $A \sim \pi(\cdot|S,\boldsymbol{\theta})$
    Take action A, observe S', R
    $\delta \leftarrow R + \gamma\hat{v}(S',\mathbf{w}) - \hat{v}(S,\mathbf{w})$ (if S' is terminal, then $\hat{v}(S',\mathbf{w}) \doteq 0$)
    $\mathbf{z}^{\mathbf{w}} \leftarrow \gamma\lambda^{\mathbf{w}}\mathbf{z}^{\mathbf{w}} + \nabla\hat{v}(S,\mathbf{w})$
    $\mathbf{z}^{\boldsymbol{\theta}} \leftarrow \gamma\lambda^{\boldsymbol{\theta}}\mathbf{z}^{\boldsymbol{\theta}} + I\nabla\ln\pi(A|S,\boldsymbol{\theta})$
    $\mathbf{w} \leftarrow \mathbf{w} + \alpha^{\mathbf{w}}\delta\mathbf{z}^{\mathbf{w}}$
    $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha^{\boldsymbol{\theta}}\delta\mathbf{z}^{\boldsymbol{\theta}}$
    $I \leftarrow \gamma I$
    $S \leftarrow S'$

Replace the gradients with traces

Off-Policy Learning with Policy Gradient Methods

- We have only talked about on-policy methods.

  This is fine since the parameterized policy can approximate the optimal deterministic policy.

- **Issue**: We cannot use data collected from a different policy!

- **Off-policy Actor-Critic methods** exist and are typically based on neural networks:
  - Deep Deterministic Policy Gradient (DDPG)
  - Twin Delayed DDPG (TD3)
  - Soft Actor Critic (SAC)

Continuous Actions

Parameterized policies can represent continuous actions!

Policy Parameterization for Continuous Actions

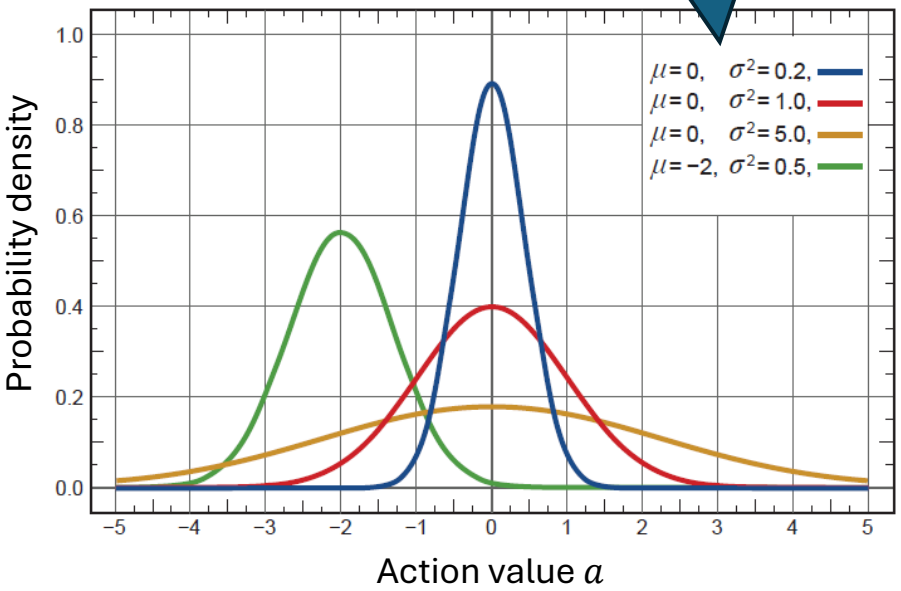
Soft-max in action preferences works for a small number of discrete actions.

For continuous actions: Choose action from a continuous distribution and learn the distribution parameters.

Example: Choose the action (value) from a Gaussian distribution:

$$\pi(a|s,\boldsymbol{\theta}) = \frac{1}{\sigma(s,\boldsymbol{\theta})\sqrt{2\pi}} \exp\left(-\frac{\left(a-\mu(s,\boldsymbol{\theta})\right)^2}{2\sigma(s,\boldsymbol{\theta})^2}\right)$$

Depend on state and learned parameters θ



where often

$$\mu(s,\boldsymbol{\theta}) = \boldsymbol{\theta}_\mu^\top \boldsymbol{x}_\mu(s)$$
$$\sigma(s,\boldsymbol{\theta}) = \exp(\boldsymbol{\theta}_\sigma^\top \boldsymbol{x}_\sigma(s))$$

Linear in state features.

is learned with $\boldsymbol{\theta} = [\boldsymbol{\theta}_\mu, \boldsymbol{\theta}_\sigma]$

What do you need to know?

- Learning a **parameterized policy** has many advantages:
  - Learn probabilities for taking an action (**stochastic policies**).
  - The policy starts typically soft and **automatically explores** and later can approximate the optimal discrete policy.
  - Can handle **continuous action** spaces.
  - **Policy gradient theorem** provides strong performance bounds.

- Important methods:
  - **Soft-max in action preferences** is a very popular choice for parameterized policies with a small set of discrete actions.
  - **REINFORCE:** Use MC to learn the policy parameters.
  - **Actor-Critic Methods:** Estimate policy and value function parameters together using TD. Reduces REINFORCE's variance and leads to more stable and faster learning.

- **Note**: Policy gradient methods (REINFORCE, actor-critic methods) have a different set of strengths and weaknesses compared to action-value methods, and the best choice depends on the problem.